there’s no limit to what you can learn about your consumers and how they use your system.

If you’re practicing semantic logging and logging custom metrics, then you can get just about any kind of data you want. Perhaps you have search functionality on your site and you want to know what people are searching for so you can better tune your content of products based on what consumers are looking for. You could save every search to the database and then do some kind of data-mining exercise on it every so often, or you could log a strongly typed event with every search.

Using the frameworks or logging that we talked about earlier in the chapter to view this data in real time makes tracking metrics that you define as success criteria for your features much easier. Let’s say that you have a gardening app where users enter the plants in their garden and search for advice on how to make them as healthy as possible. If your business model is based on creating the most valuable content for your users, then the searches people plug into your app are invaluable information because it tells you what your user base wants to read.

The key to semantic logging analysis is that it tells you what you set your system up to tell you. If you’re logging metrics that are valuable to your business and watching them as you update your software products, they become some of the most valuable metrics you can collect.

Reacting to Data in Your Logs

You should strive for zero errors in your logs. For those of you laughing right now, at least try to clean up your errors with every release. If you’re seeing large numbers of errors or warnings in your log analysis, then you have a strong data point you can use to lobby for time to clean up tech debt.

6.2.4. Tools used to collect production system data

Table 6.1 is summary of the tools we’ve talked about and used in this chapter. Keep in mind that open source systems typically have a lot of setup you have to do yourself (DIY).